PHYS 350: Honours Electricity and Magnetism

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Abstract

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1 Introduction

Textbook used is David J. Griffiths - Introduction to Electrodynamics Fifth Edition. Course grading is divided as follows:

- 35% problem sets (must be handwritten scans).
- 25% midterm.
- 40% final exam.

2 Vector Analysis

August 31 - September 4, 2026

2.1 Vector Algebra

Definition 1 (Scalar)**.** *A scalar is a quantity with just a magnitude.*

Definition 2 (Vector)**.** *A vector is a quantity with a magnitude and a direction, i.e.* $\vec{v} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z}$.

Definition 3 (Cross Product)**.** *The cross product is a vector perpendicular to both, set by the right hand rule, and it vanishes for parallel vectors:*

$$\vec{v}\times\vec{w} = |\vec{v}||\vec{w}|\sin\theta\,\hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \hat{x}(v_yw_z - w_yv_z) - \hat{y}(v_xw_z - w_xv_z) + \hat{z}(v_xw_y - w_xv_y). \tag{1}$$

Definition 4 (Dot Product)**.** *The dot product measures how much of* $\vec{v}$ *points along* $\vec{w}$*, so it is zero for perpendicular vectors:*

$$\vec{v}\cdot\vec{w} = v_xw_x + v_yw_y + v_zw_z = |\vec{v}||\vec{w}|\cos\theta. \tag{2}$$

Remark 1. *Note* $\vec{v}\times\vec{w} = -\vec{w}\times\vec{v}$*, and* $|\vec{v}| = \sqrt{\vec{v}\cdot\vec{v}}$.

Definition 5 (Field)**.** *A field is something with a value at every point in space. A scalar field assigns a number to each point. A vector field assigns a vector to each point.*

2.2 Differential Calculus

Definition 6 (Del operator)**.** *The del operator*

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right).$$

Definition 7 (Gradient)**.** *The gradient turns a scalar into a vector, pointing in the direction of steepest increase of* f*:*

$$\vec{\nabla}f = \frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y} + \frac{\partial f}{\partial z}\hat{z}. \tag{3}$$

Remark 2. *The gradient points in the steepest uphill direction, and its length grows where the hill is steeper. It is always perpendicular to the contours of constant f.*

Definition 8 (Divergence)**.** *Divergence turns a vector into a scalar, measuring sources and sinks. For a small box, non-zero divergence means the inflow and outflow through its faces do not balance: arrows spreading from a point (positive) mean fluid is created there, arrows converging (negative) mean it is destroyed.*

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \tag{4}$$

Definition 9 (Curl)**.** *Curl turns a vector into a vector, measuring local rotation. Picture a small paddle wheel: if the flow spins it, the curl is non-zero there. Uniform flow does not spin it, but shear flow (parallel arrows, longer on one side) does, so straight field lines do not imply zero curl.*

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial}{\partial y} F_z - \frac{\partial}{\partial z} F_y, -(\frac{\partial}{\partial x} F_z - \frac{\partial}{\partial z} F_x), \frac{\partial}{\partial x} F_y - \frac{\partial}{\partial y} F_x \right). \tag{5}$$

Remark 3. *The direction of the curl follows the right-hand rule: curl your fingers along the paddle wheel's rotation and your thumb points along $\vec{\nabla} \times \vec{F}$.*

Remark 4. *Divergence and curl are themselves fields: they can vanish in some places and not others, and a general field has both.*

2.3 Integral Calculus

Definition 10 (Line Integral)**.** *A line integral sums $\vec{F}$ along a curve, picking up only the component tangent to the path:*

$$\int_C \vec{F} \cdot d\vec{l}.$$

Example. *Take $\vec{F} = y\hat{x}$ along a counterclockwise quarter circle of radius R.*

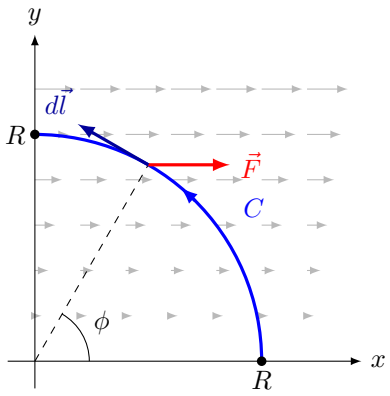


Figure 1: The field $\vec{F} = y\hat{x}$ (gray) and the quarter circle C. At a typical point, $\vec{F}$ points along $+\hat{x}$ while $d\vec{l}$ has a $-\hat{x}$ component, so $\vec{F} \cdot d\vec{l} < 0$.

On the circle $d\vec{l} = \frac{d\vec{r}}{d\phi} d\phi = R(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y})d\phi$ *and* $y = R\sin\phi$, *so*

$$\int_C \vec{F} \cdot d\vec{l} = \int_0^{\pi/2} \begin{pmatrix} R\sin\phi \\ 0 \end{pmatrix} \cdot R \begin{pmatrix} -\sin\phi \\ \cos\phi \end{pmatrix} d\phi = -R^2 \int_0^{\pi/2} \sin^2\phi\, d\phi = -\frac{R^2\pi}{4}.$$

It is negative because $\vec{F}$ *points along* $+\hat{x}$ *while we travel in the* $-\hat{x}$ *direction over most of the arc.*

Definition 11 (Flux)**.** *Flux measures how much of the field pokes through a surface:*

$$\iint_S \vec{F} \cdot d\vec{a} = \iint_S \vec{F} \cdot \hat{n}\, da.$$

Remark 5. *Flux is the total amount of a vector field flowing through a surface. Divergence is the flux per unit volume at a specific point.*

Definition 12 (Volume Integral)**.** *A volume integral sums a scalar over a region:*

$$\iiint_V f\, d\tau = \iiint_V f\, dx\, dy\, dz.$$

Remark 6. *We write* $\oint$ *for a closed loop and* $\oiint$ *for a closed surface.*

2.3.1 The Fundamental Theorems

Theorem 1 (The Fundamental Theorem of Calculus)**.** *The boundary of an interval is two points:*

$$\int_a^b f'(x)\, dx = f(b) - f(a).$$

Theorem 2 (The Fundamental Theorem for Gradients)**.**

$$\int_a^b \vec{\nabla} T \cdot d\vec{l} = T(b) - T(a).$$

The boundary of a curve is its two endpoints. To find the Eiffel Tower's height, you can sum the rise of every step (left side) or subtract altimeter readings at the top and bottom (right side).

Theorem 3 (The Fundamental Theorem for Curls)**.** *Another name for this theorem is Stokes' theorem.*

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{a} = \oint_C \vec{F} \cdot d\vec{l}. \tag{6}$$

Geometric meaning: to get the total swirl, I can work out the rotation at the boundary, or add up all the little whirlpools inside. Neighbouring whirlpools cancel along their shared edges, leaving only the boundary.

Theorem 4 (The Fundamental Theorem for Divergence)**.** *It has 2 important names: Gauss's theorem (or the divergence theorem) and Green's theorem. Respectively:*

$$\oiint_S \vec{F} \cdot d\vec{a} = \iiint_V (\vec{\nabla} \cdot \vec{F})\, d\tau,$$

$$\oint_C P dx + Q dy = \iint_R (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}) dx dy.$$

2.4 Spherical Coordinates

Definition 13 (Spherical coordinates)**.** *r is the distance from the origin, $\theta \in [0, \pi]$ the angle down from the $+z$ axis, and $\phi \in [0, 2\pi)$ the angle around it. The unit vectors $\hat{r}, \hat{\theta}, \hat{\phi}$ change direction from point to point, so they cannot be pulled out of an integral; convert to $\hat{x}, \hat{y}, \hat{z}$ first. The elements are*

$$d\vec{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}, \qquad d\tau = r^2 \sin\theta\,dr\,d\theta\,d\phi, \qquad d\vec{a}_{sphere} = R^2 \sin\theta\,d\theta\,d\phi\,\hat{r}.$$

Proposition 1 (Derivatives in spherical coordinates)**.**

$$\vec{\nabla} f = \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\hat{\phi},$$

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 F_r\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\, F_\theta\right) + \frac{1}{r\sin\theta}\frac{\partial F_\phi}{\partial \phi},$$

$$\vec{\nabla} \times \vec{F} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta\, F_\phi) - \frac{\partial F_\theta}{\partial \phi}\right]\hat{r} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial F_r}{\partial \phi} - \frac{\partial}{\partial r}(r F_\phi)\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r F_\theta) - \frac{\partial F_r}{\partial \theta}\right]\hat{\phi},$$

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}.$$

Remark 7. *For a purely radial field $\vec{F} = F_r(r)\hat{r}$, only the first term of the divergence survives, and the curl vanishes. So every central field $f(r)\hat{r}$ is curl-free.*

2.5 Index Notation

Remark 8. *A matrix carries two indices, A_{ij}, where i says which row and j says which column.*

Definition 14 (Transpose)**.** *The transpose swaps the indices, $(A^T)_{ij} = A_{ji}$.*

Definition 15 (Symmetry)**.** *A symmetric matrix has $A_{ij} = A_{ji}$, and an anti-symmetric one has $B_{ij} = -B_{ji}$.*

Definition 16 (Kronecker delta)**.** *The Kronecker delta equals $+1$ when its condition is fulfilled and 0 otherwise, so it could be the identity matrix in index notation:*

$$\delta_{ij} = \begin{cases} +1 & i = j \\ 0 & i \neq j \end{cases}.$$

Example.

$$\sum_j \delta_{ij} v_j = v_i.$$

Proposition 2 (Matrix Multiplication)**.**

$$\vec{w} = A\vec{v} \implies w_i = \sum_j A_{ij} v_j.$$

Proposition 3 (Dot Product)**.**

$$\vec{v} \cdot \vec{w} = \sum_i v_i w_i; \vec{\nabla} \cdot \vec{F} = \sum_i \partial_i F_i.$$

Proposition 4 (Cross Product)**.**

$$(\vec{v} \times \vec{w})_i = \sum_{j,k} \varepsilon_{ijk} v_j w_k; (\vec{\nabla} \times \vec{F})_i = \sum_{j,k} \varepsilon_{ijk} \partial_j F_k.$$

Definition 17 (Levi-Civita symbol)**.** *ε_{ijk} is defined by*

- $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = +1$.
- $\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1$.
- *If any two indices are equal, $\varepsilon_{ijk} = 0$.*

Example. *For $i = 1$, only the terms with $(j, k) = (2, 3)$ and $(3, 2)$ survive:*

$$(\vec{v} \times \vec{w})_1 = \varepsilon_{123} v_2 w_3 + \varepsilon_{132} v_3 w_2 = v_2 w_3 - v_3 w_2.$$

Proposition 5 (Epsilon-delta identity)**.** *Contracting two Levi-Civita symbols on a shared index gives*

$$\sum_i \varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}.$$

2.6 Second Derivatives Identities

Definition 18 (Laplacian)**.** *Taking the divergence of a gradient gives*

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \equiv \nabla^2 f, \tag{7}$$

where ∇^2 *is the Laplacian*[1]*.*

Remark 9. ∇^2 *also acts on vector fields, component by component:* $(\nabla^2 \vec{F})_i = \nabla^2 F_i$*.*

The full list of second derivative identities:

Name	**Identity**
Divergence of a gradient	$\vec{\nabla} \cdot (\vec{\nabla} f) = \nabla^2 f$
Curl of a gradient	$\vec{\nabla} \times (\vec{\nabla} f) = 0$
Divergence of a curl	$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$
Curl of a curl	$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$

Table 1: Second derivative identities.

The curl of a curl identity and the divergence of a gradient identity are proved in Appendix B.1.

2.7 The Dirac Delta Function

Definition 19 (Dirac delta)**.** $\delta(x)$ *is zero everywhere except at* $x = 0$*, where it is an infinitely tall spike of unit area:*

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1.$$

Proposition 6 (Properties)**.** *(i) Sifting:* $\int_a^b f(x)\, \delta(x - x_0)\, dx = f(x_0)$ *if* $a < x_0 < b$*, and* 0 *if* x_0 *is outside* $[a, b]$*.*

(ii) Scaling: $\delta(kx) = \frac{1}{|k|}\delta(x)$*.*

(iii) In 3D: $\iiint_V f(\vec{r})\, \delta^3(\vec{r} - \vec{r}_0)\, d\tau = f(\vec{r}_0)$ *if* $\vec{r}_0 \in V$*.*

Example. $\int_0^3 x^3 \delta(x + 1)\, dx = 0$ *since* $x = -1$ *is outside the range, while*

$$\int_{-1}^{1} 9x^2\, \delta(3x + 1)\, dx = \frac{1}{3} \int_{-1}^{1} 9x^2\, \delta\big(x + \tfrac{1}{3}\big)\, dx = \frac{1}{3} \cdot 9 \cdot \frac{1}{9} = \frac{1}{3}.$$

[1] Mathematicians often write Δ, but in physics we use ∇^2.

Theorem 5 (Divergence of $\hat{r}/r^2$)**.** *By the spherical divergence,* $\vec{\nabla}\cdot\left(\hat{r}/r^2\right) = \frac{1}{r^2}\partial_r(1) = 0$ *for* $r \neq 0$*, yet its flux through any sphere is* $\oiint \frac{1}{R^2} R^2 \sin\theta \, d\theta \, d\phi = 4\pi$*. By the divergence theorem, everything sits at the origin:*

$$\vec{\nabla}\cdot\left(\frac{\hat{r}}{r^2}\right) = 4\pi\,\delta^3(\vec{r}).$$

So for a point charge, $\vec{\nabla}\cdot\vec{E} = \frac{q}{\epsilon_0}\delta^3(\vec{r}) = \rho/\epsilon_0$*, as Gauss' law requires.*

2.8 The Theory of Vector Fields

Remark 10. *The condition at infinity is needed: a field can be both curl-free and divergence-free without being zero, e.g.* $\vec{F} = yz\,\hat{x} + zx\,\hat{y} + xy\,\hat{z} = \vec{\nabla}(xyz)$*. Adding it to any field changes neither* $\vec{\nabla}\cdot\vec{F}$ *nor* $\vec{\nabla}\times\vec{F}$*.*

Theorem 6 (Irrotational fields)**.** *Let* $\vec{F}$ *be a* C^1 *vector field on a simply connected domain (e.g. all of* $\mathbb{R}^3$*). The following are equivalent:*

(i) $\vec{\nabla}\times\vec{F} = 0$ *everywhere.*

(ii) $\int_a^b \vec{F}\cdot d\vec{l}$ *is independent of path, for any given end points.*

(iii) $\oint_C \vec{F}\cdot d\vec{l} = 0$ *for every closed loop* C*.*

(iv) $\vec{F} = -\vec{\nabla}\psi$ *for some scalar field* ψ*.*

Theorem 7 (Solenoidal fields)**.** *Let* $\vec{F}$ *be a* C^1 *vector field on a domain with no enclosed voids (e.g. all of* $\mathbb{R}^3$*). The following are equivalent:*

(i) $\vec{\nabla}\cdot\vec{F} = 0$ *everywhere.*

(ii) $\iint_S \vec{F}\cdot d\vec{a}$ *is independent of surface, for any given boundary line.*

(iii) $\oiint_S \vec{F}\cdot d\vec{a} = 0$ *for any closed surface* S*.*

(iv) $\vec{F} = \vec{\nabla}\times\vec{A}$ *for some vector field* $\vec{A}$*.*

Remark 11. *In electrostatics we will find* $\vec{\nabla}\times\vec{E} = 0$*, which licenses* $\vec{E} = -\vec{\nabla}V$ *and the electric potential. In magnetostatics we will find* $\vec{\nabla}\cdot\vec{B} = 0$*, which licenses* $\vec{B} = \vec{\nabla}\times\vec{A}$*, the vector potential.*

Theorem 8 (Helmholtz)**.** *If* $\vec{\nabla}\cdot\vec{F}$ *and* $\vec{\nabla}\times\vec{F}$ *are specified everywhere and* $\vec{F}\to 0$ *at infinity, then* $\vec{F}$ *is uniquely determined. It can be written as* $\vec{F} = \vec{\nabla}\psi + \vec{\nabla}\times\vec{A}$*, an irrotational part plus a solenoidal part.*

3 Electrostatics

September 11-16 & 21-25, 2026

Definition 20 (Electrostatics)**.** *Electrostatics is the study of stationary electric charges.*

3.1 Electric Field

Goal of electrodynamics: given source charges with known positions, find the force on a test charge Q.

Proposition 7 (Principle of Superposition)**.** *The force between two charges is unaffected by other charges, so*

$$\vec{F} = \sum_i \vec{F}_i.$$

Law 1 (Coulomb's law)**.** *The force on a test charge Q due to a point charge q is*

$$\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{qQ}{\mathfrak{r}^2}\,\hat{\mathfrak{r}}\,[N]\,, \tag{8}$$

where $\vec{\mathfrak{r}} = \vec{r} - \vec{r'}$ is the separation vector from the location of q ($\vec{r'}$) to Q ($\vec{r}$).

Remark 12. *Coulomb's law has the same shape as Newton's gravity. The reason is geometric rather than a coincidence, and Gauss' law makes that explicit: in d spatial dimensions the flux spreads over a $(d-1)$-dimensional sphere, so $F \propto 1/\mathfrak{r}^{d-1}$, and the 4π is the area of the unit sphere in $d = 3$.*

Definition 21 (Permittivity of free space)**.** *The permittivity of free space measures how easily an electric field can form in a classical vacuum:*

$$\epsilon_0 = 8.85 \times 10^{-12}\ \left[\frac{C^2}{N \cdot m^2}\right].$$

By the principle of superposition, the total force on a test charge Q due to several point charges is

$$\vec{F} = \frac{Q}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{\mathfrak{r}_i^2}\,\hat{\mathfrak{r}}_i.$$

Which leads us to the definition of the electric field

$$\vec{F} = Q\vec{E}.$$

Definition 22 (Electric field)**.** *The electric field is the force per unit test charge.*

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{\mathfrak{r}_i^2}\,\hat{\mathfrak{r}}_i. \tag{9}$$

$\vec{\mathfrak{r}}_i \equiv \vec{r} - \vec{r}_i$, where $\vec{r}$ is the location of the field point, where we want to know what force a charge would feel, and $\vec{r}_i$ is the location of the source charge q_i.

3.1.1 Continuous Charge Distributions

Definition 23 (Continuous distributions)**.** *For a charge along a line, over a surface, or through a volume, replace q_i by $dq = \lambda\, dl' = \sigma\, da' = \rho\, d\tau'$ and the sum becomes an integral:*

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{\mathfrak{r}^2}\,\hat{\mathfrak{r}}, \qquad \vec{\mathfrak{r}} = \vec{r} - \vec{r}', \tag{10}$$

where $\vec{r}$ is the field point and $\vec{r}'$ the source point.

Example (Line of charge)**.** *A straight segment of length $2L$, centred at the origin on the x axis, carries a uniform line charge density λ. Find $\vec{E}$ at the point $P = (0, 0, z)$, a height z above its midpoint.*

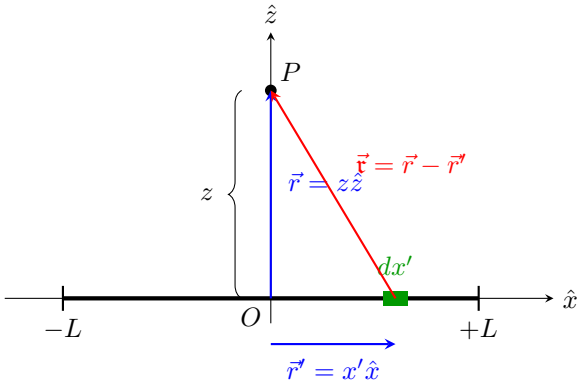


Figure 2: Setup for the finite line of charge. The source point $\vec{r}' = x'\hat{x}$ runs from $-L$ to L; the field point is P at $\vec{r} = z\hat{z}$; the separation vector is $\vec{\mathfrak{r}} = \vec{r} - \vec{r}'$.

With $\vec{r} = z\hat{z}$ and $\vec{r}' = x'\hat{x}$,

$$\vec{\mathfrak{r}} = z\hat{z} - x'\hat{x}, \qquad \mathfrak{r} = \sqrt{z^2 + x'^2},$$

so

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0}\int_{-L}^{L} \frac{z\hat{z} - x'\hat{x}}{(z^2 + x'^2)^{3/2}}\,dx' = \frac{\lambda}{4\pi\epsilon_0}\left[\hat{z}\int_{-L}^{L}\frac{z}{(z^2 + x'^2)^{3/2}}dx' - \hat{x}\int_{-L}^{L}\frac{x'}{(z^2 + x'^2)^{3/2}}dx'\right].$$

The second integral vanishes by symmetry: its integrand is odd in x', so the pieces at x' and $-x'$ cancel. The first is done with the substitution $x' = z\tan\theta$, giving

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0}\,\frac{1}{z}\left[\frac{x'}{\sqrt{x'^2 + z^2}}\right]_{-L}^{L}\hat{z} = \frac{\lambda}{4\pi\epsilon_0}\,\frac{2L}{z\sqrt{L^2 + z^2}}\,\hat{z} = \frac{\lambda}{2\pi\epsilon_0 z}\cdot\frac{L}{\sqrt{L^2 + z^2}}\,\hat{z}.$$

This has two limits:

- *Close to the line ($z \ll L$), it looks like an infinite line:* $\vec{E} \to \dfrac{\lambda}{2\pi\epsilon_0 z}\hat{z}$.
- *Far from the line ($z \gg L$), it looks like a point charge:* $\vec{E} \to \dfrac{Q}{4\pi\epsilon_0 z^2}\hat{z}$ *with* $Q = 2\lambda L$.

3.2 Divergence and Curl of Electrostatic Fields

3.2.1 Gauss's Law

Law 2 (Gauss' law)**.** *The flux of $\vec{E}$ through any closed surface is set by the charge inside it:*

$$\oiint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}. \tag{11}$$

Here Q_{enc} is the total charge of the object inside the gaussian surface S.

Proposition 8 (Differential form)**.** *Apply the divergence theorem (Theorem 4) to the left side and write $Q_{enc} = \iiint_V \rho\, d\tau$:*

$$\iiint_V (\vec{\nabla} \cdot \vec{E})\, d\tau = \frac{1}{\epsilon_0} \iiint_V \rho\, d\tau,$$

where ρ is the charge density. Since this holds for every volume V, the integrands match:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}. \tag{12}$$

Definition 24 (Gaussian Surface)**.** *The Gaussian surface just means an imaginary surface (nothing is physically there).*

Remark 13. *If there is no charge inside the Gaussian surface, the net flux is zero: every field line that enters on one side leaves on another.*

3.2.2 Applications of Gauss's Law

Gauss's law always holds but is only useful with symmetry: we need $\vec{E}$ parallel to $d\vec{a}$ and $|\vec{E}|$ constant over the Gaussian surface to pull $|\vec{E}|$ out of the integral. Only three symmetries work: spherical, cylindrical, and planar.

Spherical Symmetry.

Example (Uniform solid sphere)**.** *Let a uniform solid sphere of radius R and total charge q. Imagine a gaussian sphere of radius r, so*

$$E \oiint da = \frac{Q_{enc}}{\epsilon_0},$$

let $Q_{enc} = q\,(r^3/R^3)$ *for* $r < R$ *and* $Q_{enc} = q$ *for* $r > R$. *The area is* $\oiint da = 4\pi r^2$, *so*

$$\vec{E} = \begin{cases} \dfrac{q\,r}{4\pi\epsilon_0 R^3}\hat{r} & r < R \\ \frac{q}{4\pi\epsilon_0 R^2}\hat{r} & r = R \\ \dfrac{q}{4\pi\epsilon_0 r^2}\hat{r} & r > R \end{cases}.$$

Example. *A thin shell instead gives*

$$\vec{E} = \begin{cases} 0 & r < R \\ \frac{\sigma}{\epsilon_0}\hat{r} & r = R \\ \dfrac{q}{4\pi\epsilon_0 r^2}\hat{r} & r > R \end{cases}.$$

Remark 14. *A uniformly charged shell that pulses in and out,* $R(t) = R_0 + A\sin\omega t$, *keeps the constant field* $\frac{Q}{4\pi\epsilon_0 r^2}\hat{r}$ *outside: accelerating charges, but nothing escapes to infinity, so no radiation.*

Cylindrical Symmetry.

Example (Infinite line of charge)**.** *For a line with charge* λ *per unit length, take the Gaussian surface to be a coaxial cylinder of radius* z *(the distance from the line to the field point, not the radius of the line) and length* d. $\vec{E}$ *points radially outward, so* $\vec{E}\cdot d\vec{a} = 0$ *on the end caps and only the curved side contributes:*

$$E\,(2\pi z d) = \frac{\lambda d}{\epsilon_0} \implies E = \frac{\lambda}{2\pi\epsilon_0 z}.$$

Example (Finite versus infinite line)**.** *Near its midpoint, a finite line has a weaker field than an infinite one: the same enclosed charge, but some flux leaks through the caps.*

Problem 1 (Midterm 2025, Q2)**.** *An infinite non-conducting tube along the* x *axis has* $\rho(s) = A\,s/a$ *for* $s < a$ *and* 0 *elsewhere, with* s *the distance from the axis. Find* $\vec{E}$ *everywhere.*

Planar Symmetry.

Example (Infinite plane of charge)**.** *For a plane with surface charge density* σ, *take the Gaussian surface to be a pillbox straddling the plane, with caps of area* A. $\vec{E}$ *points away from the plane on both sides, so there is no flux through the side wall and* EA *through each cap:*

$$2EA = \frac{\sigma A}{\epsilon_0} \implies E = \frac{\sigma}{2\epsilon_0}.$$

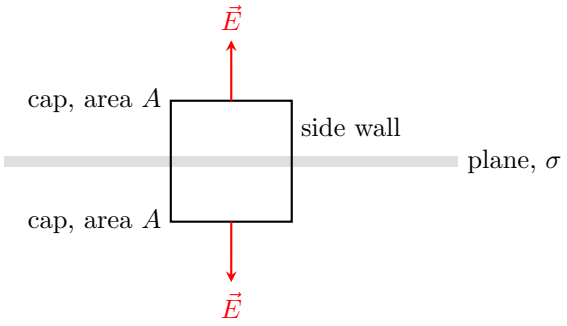


Figure 3: Side view of the pillbox straddling the plane.

Example (Two planes)**.** *Superpose two sheets a distance d apart, each giving $\sigma/2\epsilon_0$, pointing away from a positive sheet and toward a negative one.*

- *$+\sigma$ and $-\sigma$ (parallel plates):*

$$E = \begin{cases} \frac{\sigma}{\epsilon_0} & \text{between the plates,} \\ 0 & \text{outside the plates.} \end{cases}$$

- *$+\sigma$ and $+\sigma$:*

$$E = \begin{cases} 0 & \text{between the plates,} \\ \frac{\sigma}{\epsilon_0} & \text{outside the plates.} \end{cases}$$

Remark 15 (Fall-off)**.** *The fall-off is how fast E decreases with the distance r from the source: $E \propto 1/r^2$ for a point charge, $1/r$ for an infinite line, and r^0 (uniform) for an infinite plane. To see why, picture a narrow cone opening from the field point toward the source. Its cross section at distance r grows like r^2, which alone gives the $1/r^2$ of a point charge, but for extended sources the charge inside the cone grows too: it intercepts a length $\propto r$ of a line, giving $E \propto r/r^2 = 1/r$, and an area $\propto r^2$ of a sheet, giving $E \propto r^2/r^2$, a constant.*

Remark 16. *For two equal positive charges, no field lines cross the midplane: the field there is parallel to the plane and vanishes at the midpoint. Gauss's law still holds, but without symmetry it cannot solve for $\vec{E}$.*

3.2.3 The Curl of $\vec{E}$

Theorem 9 (Electrostatic fields are curl-free)**.** *For any electrostatic field,*

$$\oint \vec{E} \cdot d\vec{l} = 0, \qquad \vec{\nabla} \times \vec{E} = 0. \tag{13}$$

Proof. Let there be a point charge at the origin. The field is radial, $\vec{E} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$, so the line integral from some point a to some other point b in three dimensions

$\int_a^b \vec{E} \cdot d\vec{l}$, in spherical coordinates $d\vec{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$, we get

$$\vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0 r^2} dr.$$

Therefore,

$$\int_a^b \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right).$$

The integral around a closed path is evidently zero (for then $r_a = r_b$):

$$\oint \vec{E} \cdot d\vec{l} = 0,$$

and hence, by Stokes' theorem (6), $\vec{\nabla} \times \vec{E} = 0$. □

3.3 Electric Potential

Definition 25 (Electric potential)**.** *Since $\vec{\nabla} \times \vec{E} = 0$, the irrotational fields theorem 6 guarantees a scalar V with*

$$\vec{E} = -\vec{\nabla} V. \tag{14}$$

The minus sign is convention: positive charges are pushed toward lower potential energy.

Proposition 9 (Potential from the field)**.**

$$V(\vec{r}) = -\int_{ref}^{\vec{r}} \vec{E} \cdot d\vec{l} \left[\frac{J}{C} = volts\ (v) \right]. \tag{15}$$

The path does not matter since the integral is path-independent[2]*. By convention the reference is at infinity, unless the charge distribution extends to infinity.*

Proposition 10 (Potential of a Charge Distribution Extending to Infinity)**.** *If the charge distribution extends to infinity, the reference point for the potential is typically taken at a finite distance where the potential vanishes, for example a point on the plane.*

Definition 26 (Equipotential)**.** *An equipotential is a region, typically a surface, on which V is constant.*

Example (Point charge)**.** *Along a radial path,*

$$V(r) = -\frac{q}{4\pi\epsilon_0} \int_\infty^r \frac{dr'}{r'^2} = \frac{q}{4\pi\epsilon_0 r}.$$

[2] Changing the reference point adds a constant to V, which leaves $\vec{E} = -\vec{\nabla} V$ unchanged (gauge freedom). Statement (ii) of the irrotational fields theorem.

Example (Uniform solid sphere)**.** *Using the field of Example 3.2.2, outside* $V = \frac{q}{4\pi\epsilon_0 r}$*. Inside, integrate in from infinity through both regions:*

$$V(r) = -\int_\infty^R \frac{q\,dr'}{4\pi\epsilon_0 r'^2} - \int_R^r \frac{q\,r'\,dr'}{4\pi\epsilon_0 R^3} = \frac{q}{8\pi\epsilon_0 R}\left(3 - \frac{r^2}{R^2}\right).$$

Problem 2 (Midterm drills)**.** *Find* V *(i) at the centre of a ring of radius* a *and charge* Q*; (ii) at* $x = 2a$ *for a line of charge* Q *from* $x = -a$ *to* $x = a$*.*

3.3.1 Poisson's Equation and Laplace's Equation

Proposition 11 (Poisson's equation)**.** *Combining* $\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0$ *with* $\vec{E} = -\vec{\nabla}V$*,*

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}. \tag{16}$$

Corollary 9.1 (Laplace's equation)**.** *In regions where there is no charge, so* $\rho = 0$*, Poisson's equation reduces to Laplace's equation:*

$$\nabla^2 V = 0. \tag{17}$$

3.3.2 The Potential of a Localized Charge Distribution

Proposition 12 (Potential of a distribution)**.** *By superposition (7),*

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0}\iiint \frac{\rho(\vec{r}')}{\mathfrak{r}}\,d\tau', \tag{18}$$

where $\rho(\vec{r}')d\tau' = dq$*. Unlike the integral for* $\vec{E}$*, there is no* $\hat{\mathfrak{r}}$*, so computing* V *first and then* $\vec{E} = -\vec{\nabla}V$ *is often easier.*

Remark 17 (The six relations between ρ, $\vec{E}$ and V)**.** *Any one of the three determines the other two:*

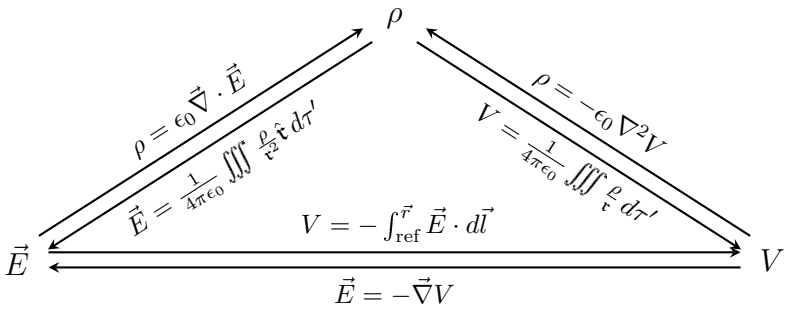


Figure 4: The six relations of electrostatics. In practice ρ is the source, $\vec{E}$ is what we measure, and V is the easiest to compute.

3.3.3 Boundary Conditions

A surface charge σ makes $\vec{E}$ jump as you cross it. To fix this we use Gauss's law.

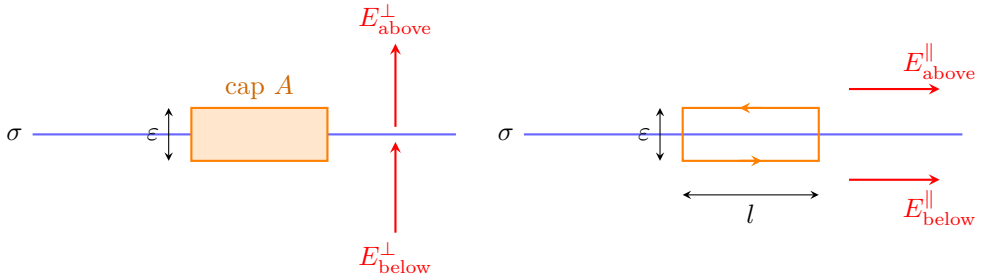


Figure 5: Side views of a charged surface.

Proposition 13 (Normal component)**.** *Take a pillbox of cap area A straddling the surface and shrink its thickness $\varepsilon \to 0$, so the side wall carries no flux. Only the caps count, and the charge inside is σA:*

$$E^{\perp}_{above}A - E^{\perp}_{below}A = \frac{\sigma A}{\epsilon_0} \quad \Longrightarrow \quad E^{\perp}_{above} - E^{\perp}_{below} = \frac{\sigma}{\epsilon_0}.$$

Proposition 14 (Tangential component)**.** *Take a thin rectangular loop of length l straddling the surface, again with $\varepsilon \to 0$ so the short sides contribute nothing. Since $\oint \vec{E} \cdot d\vec{l} = 0$,*

$$E^{\parallel}_{above}\, l - E^{\parallel}_{below}\, l = 0 \quad \Longrightarrow \quad \vec{E}^{\parallel}_{above} = \vec{E}^{\parallel}_{below}.$$

Theorem 10 (Boundary condition on $\vec{E}$)**.** *With $\hat{n}$ the unit normal pointing from below to above, the two results combine into*

$$\vec{E}_{above} - \vec{E}_{below} = \frac{\sigma}{\epsilon_0}\hat{n}. \tag{19}$$

The normal component jumps by σ/ϵ_0; the tangential component is continuous.

Remark 18. *Check with the infinite plane: $\vec{E} = +\frac{\sigma}{2\epsilon_0}\hat{n}$ above and $-\frac{\sigma}{2\epsilon_0}\hat{n}$ below, a jump of exactly σ/ϵ_0.*

Theorem 11 (Boundary conditions on V)**.** *The potential is continuous across any boundary, since*

$$V_{above} - V_{below} = -\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l} \overset{path\ length\to 0}{\Longrightarrow} V_{above} = V_{below}.$$

However, the gradient of V is not continuous: substituting $\vec{E} = -\vec{\nabla}V$ into Theorem 10 gives

$$\vec{\nabla}V_{above} - \vec{\nabla}V_{below} = -\frac{\sigma}{\epsilon_0}\hat{n}. \tag{20}$$

3.4 Work and Energy in Electrostatics

Remark 19. *The energy of a charge configuration is the work needed to assemble it, bringing each charge in from infinity. Since $\vec{\nabla} \times \vec{E} = 0$, this work is path-independent, so it depends only on the final arrangement and is stored as potential energy. This is why W denotes both.*

Proposition 15 (Potential of a test charge)**.** *Bringing Q in from infinity against the field takes*

$$W = -Q \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{l} = QV(\vec{r}). \tag{21}$$

3.4.1 The Energy of a Point Charge Distribution

Proposition 16 (Energy of N point charges)**.** *The first charge costs no work, since no field exists yet, and each later one feels the potential $V(\vec{r}_i)$ of those already placed, giving one term per pair:*

$$W = \frac{1}{2} \sum_{i=1}^{n} q_i \left(\sum_{j \neq i}^{n} \frac{q_j}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|} \right) = \frac{1}{2} \sum_{i=1}^{n} q_i V(\vec{r}_i), \tag{22}$$

where the $\frac{1}{2}$ corrects for counting each pair twice.

3.4.2 The Energy of a Continuous Charge Distribution

Theorem 12 (Energy stored in the field)**.** *For a volume charge distribution, the energy is*

$$W = \frac{1}{2} \iiint \rho V \, d\tau = \frac{\epsilon_0}{2} \iiint_{\text{all space}} E^2 \, d\tau. \tag{23}$$

3.4.3 Comments on Electrostatic Energy

Definition 27 (Energy Density)**.** *From Theorem 12, the energy per unit volume stored in the field is*

$$u = \frac{\epsilon_0}{2} E^2. \tag{24}$$

The proof is in Appendix B.2.

Proposition 17 (Energy does not obey superposition)**.** *The energy is quadratic in the field, so for $\vec{E} = \vec{E}_1 + \vec{E}_2$,*

$$W = W_1 + W_2 + \epsilon_0 \iiint \vec{E}_1 \cdot \vec{E}_2 \, d\tau. \tag{25}$$

Proof. $E^2 = (\vec{E}_1 + \vec{E}_2) \cdot (\vec{E}_1 + \vec{E}_2) = E_1^2 + E_2^2 + 2\vec{E}_1 \cdot \vec{E}_2$. Multiply by $\frac{\epsilon_0}{2}$ and integrate over all space. □

Remark 20. *The cross term is the interaction energy of the two distributions, and it is negative when they attract. For example, doubling every charge doubles $\vec{E}$ but quadruples W.*

Problem 3 (Midterm drill)**.** *Two plates of area A, a distance d apart, carry $+\sigma$ and $-\sigma$ ($A \gg d^2$). Find the energy.*

Problem 4 (Problem Set 4, Q4)**.** *Concentric shells of radii $a < b$ carry $+q$ and $-q$. Find the energy.*

3.5 Conductors

Definition 28 (Insulator)**.** *In an insulator, such as glass or ruber, each electron is on a short leash, attached to a particular atom.*

Definition 29 (Conductor)**.** *A conductor is a material with charges that are free to move (in a metal, the electrons).*

Proposition 18. *In an external field, a conductor's free charges move to its surface until their field $\vec{E}_{int}$ cancels $\vec{E}_{ext}$ inside. In practice, it's the negative charges (the electrons) that do the moving, leaving a positive charge on the far side.*

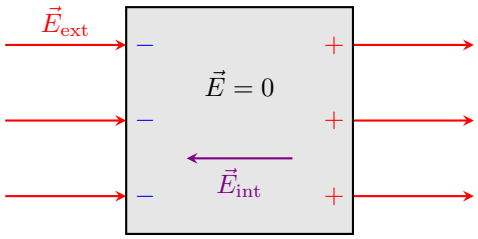


Figure 6: A conductor in an external field.

Proposition 19 (Properties of a conductor in equilibrium)**.**

(i) $\vec{E} = 0$ *inside, otherwise the free charges would still be moving.*

(ii) $\rho = 0$ *inside, since* $\rho = \epsilon_0 \vec{\nabla} \cdot \vec{E} = \epsilon_0 \cdot 0 = 0$.

(iii) Any net charge therefore sits on the surface.

(iv) A conductor is an equipotential: for any two points within (or at the surface of) a conductor, $V(\vec{b}) - V(\vec{a}) = -\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l} = 0$.

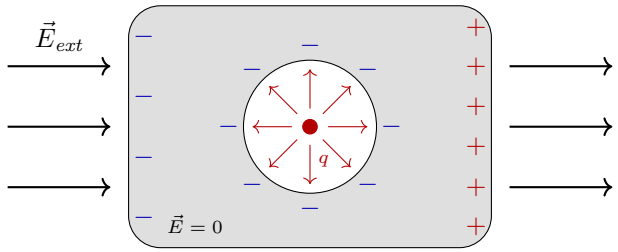
(v) Just outside, $\vec{E}$ is perpendicular to the surface. A tangential part would push the surface charges along the surface, and they would not be in equilibrium.

Remark 21. *Grounding a conductor connects it to $V(\infty) = 0$, so its whole body, not just its surface, sits at $V = 0$. Positive charge flows to the ground if the conductor would otherwise be at $V > 0$, or from the ground if it would be at $V < 0$, until $V = 0$.*

3.5.1 Induced Charges

Remark 22 (Induced Charges)**.** *If a conductor has a cavity with a charge q inside, q induces $-q$ on the cavity wall and $+q$ on the outer surface (for a neutral conductor), and the outside cannot tell how the charge inside is arranged. In particular, for a sphere the outer charge spreads uniformly even if the cavity is off centre, so the field outside is that of a point charge at the centre of the sphere, not of the cavity.*

Remark 23 (Induced Charges with External Field)**.** *Same setup as the previous remark, but now add an external field: the charge on the cavity wall is still $-q$ and the outer surface still carries $+q$, but the external field induces a nonuniform distribution on the outer surface. The field inside the cavity is unchanged, since it is due only to q and the wall charge.*



Remark 24 (Faraday Cage)**.** *If a conductor has an empty cavity, the field inside the cavity is zero: its wall is an equipotential, and a solution of Laplace's equation with constant boundary values is constant (no maxima or minima, see the Laplace's equation section).*

Example (Neutral conductor near a charged rod)**.** *A neutral metal block sits near an insulating rod with uniform positive charge. Negative charge is induced on the side facing the rod and an equal positive charge on the far side. The negative charges are closer, and Coulomb's force falls off with distance, so the attraction beats the repulsion: the neutral conductor is pulled toward the rod.*

Problem 5 (Midterm 2025, Q3)**.** *A thick conducting shell (inner radius a, outer radius b, empty cavity) has net charge $-Q$. (a) Where is the charge? (b) Sketch $E_r(r)$. (c) Sketch $V(r)$ with $V(\infty) = 0$. (d) A charge $+q$ with $q > Q$ is placed outside: sketch the negative charge. (e) Find the total charge on the inner surface and (f) on the outer surface.*

3.5.2 Surface Charge

Proposition 20 (Field at the surface of a conductor)**.** *Theorem 10 with $\vec{E}_{below} = 0$ (inside the metal) gives, just outside,*

$$\vec{E} = \frac{\sigma}{\epsilon_0}\hat{n}.$$

Remark 25. *In general, charge on a conductor concentrates where the surface is most sharply curved (edges, corners, points). Since $E = \sigma/\epsilon_0$ there, the field is strongest at sharp points, which is how lightning rods work.*

Proposition 21 (Force on a surface charge)**.** *The force per unit area on a surface charge σ is*

$$\vec{f} = \sigma\vec{E}_{avg} = \frac{\sigma}{2}\left(\vec{E}_{above} + \vec{E}_{below}\right). \tag{26}$$

For a conductor, $\vec{E}_{below} = 0$ and $\vec{E}_{above} = \frac{\sigma}{\epsilon_0}\hat{n}$, so $\vec{f} = \frac{\sigma^2}{2\epsilon_0}\hat{n}$: an outward electrostatic pressure which, in terms of the field just outside, is

$$P = \frac{\epsilon_0}{2}E^2. \tag{27}$$

Proof. Split the field at a small patch of the surface into the patch's own field and $\vec{E}_{\text{other}}$, due to everything else. The patch cannot push on itself, so only $\vec{E}_{\text{other}}$ acts on it. Up close the patch looks like an infinite plane, giving $\pm\frac{\sigma}{2\epsilon_0}\hat{n}$ on either side, so

$$\vec{E}_{\text{above}} = \vec{E}_{\text{other}} + \frac{\sigma}{2\epsilon_0}\hat{n}, \qquad \vec{E}_{\text{below}} = \vec{E}_{\text{other}} - \frac{\sigma}{2\epsilon_0}\hat{n}.$$

$\vec{E}_{\text{other}}$ is continuous across the patch, so adding the two gives $\vec{E}_{\text{other}} = \frac{1}{2}\left(\vec{E}_{\text{above}} + \vec{E}_{\text{below}}\right)$, and $\vec{f} = \sigma\vec{E}_{\text{other}}$. □

Remark 26. *Since σ enters squared, the pressure points outward whatever the sign of the charge: a charged conductor is pulled into the field. Note also that P equals the energy density u just outside.*

3.5.3 Capacitors

Definition 30 (Capacitor)**.** *A capacitor is a device that stores electrical energy in an electric field. It typically consists of two conductors separated by an insulating material (dielectric).*

Definition 31 (Capacitance)**.** *The capacitance C of a capacitor is the ratio of the charge Q on one conductor to the potential difference V between the conductors:*

$$C \equiv \frac{Q}{V}. \tag{28}$$

Proposition 22 (Work done to charge a capacitor)**.** *The work done to charge a capacitor is*

$$W = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV. \tag{29}$$

Proof. Charge the capacitor by moving charge from the negative plate to the positive one. When the plates hold $\pm q$, the potential difference is q/C, so moving an extra dq costs

$$dW = \left(\frac{q}{C}\right) dq.$$

Integrating from $q = 0$ to $q = Q$,

$$W = \frac{1}{C}\int_0^Q q\, dq = \frac{Q^2}{2C},$$

and $Q = CV$ gives the other two forms. □

4 Potentials

September 18-25, 2026

5 Magnetic Fields in Matter

6 Electrodynamics

A Extra Material

Remark 27 (Compactified dimensions)**.** *Suppose an extra dimension exists but is curled up, the way a flat sheet with coordinates (x, y) can be rolled into a cylinder whose second coordinate w is periodic with a small radius. At distances much smaller than that radius, field lines spread out into all the dimensions and the field falls off faster than $1/r^2$. At distances much larger than it, there is only room to spread out in the large dimensions and the usual $1/r^2$ is recovered. So a measured deviation from the inverse square law at short range would be evidence for extra dimensions.*

This has been tested for gravity by the Eöt-Wash torsion balance experiments (2004, 2007, 2020), which checked the $1/r^2$ law down to separations of $52\,\mu$m. Newtonian gravity fit the data, ruling out gravitational strength Yukawa interactions with ranges above $38.6\,\mu$m at 95% confidence.

B Proofs

B.1 Second Derivatives Identities Proofs

Proof. The curl of a curl. Set $\vec{G} \equiv \vec{\nabla} \times (\vec{\nabla} \times \vec{F})$. Then,

$$G_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \sum_{m,n} \varepsilon_{kmn} \partial_m F_n = \sum_{j,k,m,n} \varepsilon_{ijk} \varepsilon_{kmn} \partial_j \partial_m F_n.$$

Using the cyclic permutation $\varepsilon_{ijk} = \varepsilon_{kij}$, so that the summed index k sits first in both epsilons, the epsilon-delta identity 5 gives

$$\sum_k \varepsilon_{kij} \varepsilon_{kmn} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}.$$

Reading off the proposition with the summed index k removed:

$$G_i = \sum_{j,m,n} (\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm})\, \partial_j \partial_m F_n.$$

Each Kronecker delta collapses one sum by setting its indices equal. In the first term δ_{im} forces $m = i$ and δ_{jn} forces $n = j$; in the second, δ_{in} forces $n = i$ and δ_{jm} forces $m = j$:

$$G_i = \sum_j \partial_j \partial_i F_j - \sum_j \partial_j \partial_j F_i \overset{3}{=} \partial_i \sum_j \partial_j F_j - \sum_j \partial_j \partial_j F_i = \left(\vec{\nabla}(\vec{\nabla} \cdot \vec{F})\right)_i - (\nabla^2 \vec{F})_i.$$

$$\therefore \vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}.$$

□

[3] This equality uses $\partial_j \partial_i = \partial_i \partial_j$, valid for $\vec{F}$ twice continuously differentiable.

Proof. Curl of a gradient. Let f be twice continuously differentiable. With $(\vec{\nabla} f)_k = \partial_k f$,

$$\Big(\vec{\nabla} \times (\vec{\nabla} f)\Big)_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f.$$

The indices j and k are summed over, so they can be renamed $j \leftrightarrow k$:

$$\Big(\vec{\nabla} \times (\vec{\nabla} f)\Big)_i = \sum_{j,k} \varepsilon_{ikj} \partial_k \partial_j f.$$

Now use antisymmetry, $\varepsilon_{ikj} = -\varepsilon_{ijk}$, and the fact that partial derivatives commute, $\partial_k \partial_j f = \partial_j \partial_k f$:

$$\Big(\vec{\nabla} \times (\vec{\nabla} f)\Big)_i = -\sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f = -\Big(\vec{\nabla} \times (\vec{\nabla} f)\Big)_i .$$

A number equal to its own negative is zero, so every component vanishes and $\vec{\nabla} \times (\vec{\nabla} f) = \vec{0}$. □

B.2 Energy Stored in the Field

Proof. Start from the energy of a continuous distribution $W = \frac{1}{2} \iiint \rho V \, d\tau$ and replace ρ with Poisson's equation (16):

$$W = -\frac{\epsilon_0}{2} \iiint V \nabla^2 V \, d\tau.$$

Rewrite $V\nabla^2 V$ with the product rule $\vec{\nabla} \cdot (f\vec{A}) = \vec{\nabla} f \cdot \vec{A} + f\, \vec{\nabla} \cdot \vec{A}$, taking $f = V$ and $\vec{A} = \vec{\nabla} V$:

$$\vec{\nabla} \cdot (V \vec{\nabla} V) = |\vec{\nabla} V|^2 + V \nabla^2 V \quad \Longrightarrow \quad V \nabla^2 V = \vec{\nabla} \cdot (V \vec{\nabla} V) - |\vec{\nabla} V|^2.$$

Substituting, the minus sign in front flips both terms:

$$W = \frac{\epsilon_0}{2} \iiint |\vec{\nabla} V|^2 \, d\tau - \frac{\epsilon_0}{2} \iiint \vec{\nabla} \cdot (V \vec{\nabla} V) \, d\tau.$$

The second term is the volume integral of a divergence, so the divergence theorem turns it into a flux through the boundary surface:

$$W = \frac{\epsilon_0}{2} \iiint |\vec{\nabla} V|^2 \, d\tau - \frac{\epsilon_0}{2} \oiint V \vec{\nabla} V \cdot d\vec{a}.$$

Take the surface to be a sphere of radius $R \to \infty$. For a localized charge, $V \sim 1/R$ and $|\vec{\nabla} V| \sim 1/R^2$, while the area grows as R^2, so the surface term goes as $1/R \to 0$. With $\vec{E} = -\vec{\nabla} V$,

$$W = \frac{\epsilon_0}{2} \iiint_{\text{all space}} E^2 \, d\tau.$$

This gives the total, not where it sits. For that, pull one plate of a capacitor ($\pm\sigma$, area A, $E = \sigma/\epsilon_0$ between) a distance dx against the pressure (Proposition 21):

$$dW = PA\,dx = \frac{\epsilon_0}{2}E^2\,(A\,dx).$$

E is unchanged, so this work only fills a new volume $A\,dx$ with field:

$$u = \frac{\epsilon_0}{2}E^2.$$

□

C Useful Links

- Laplace relaxation demo:
  https://claude.ai/artifact/J2wPx8V3ajec8vDLsARnKX
- Electrostatics simulation:
  https://icphysweb.z13.web.core.windows.net/simulation.html